14 Energy and Momentum of the Electromagnetic Field

14.1 Motivation: Energy Transfer

In elementary mechanics we often use a force to change a particles energy and momentum without asking whether, if we include the agent determining the force, momentum and energy are conserved overall.

In this section, in the context of electromagnetism, we will keep track of the momentum and energy of both particles and the electromagnetic field. We will find overall conservation, reflecting when an electric field accelerates a particle, the field loses energy and momentum while the particle gains energy and momentum.

This explicit discussion of both fields and particles also resolves issues associated with Newton's third law when the momentum residing in fields is not considered. (E.g. see discussion in section 8.2 of *Introduction to Electrodynamics*, Griffiths D.J., fifth edition (electronic), CUP (2024))

To put this on a firm basis, examine (we assume no media here – see problems (3.3) and (3.11) for media):

$$\mathbf{E}\cdot(\nabla\times\mathbf{B}) = \mathbf{E}\cdot\left(\frac{1}{\mathrm{c}}\frac{\partial\mathbf{E}}{\partial t} + \frac{4\pi}{\mathrm{c}}\mathbf{j}\right) \ ; \tag{14.1}$$

$$\mathbf{B}\cdot(\nabla\times\mathbf{E}) = \mathbf{B}\cdot\left(-\frac{1}{\mathrm{c}}\frac{\partial\mathbf{B}}{\partial t}\right) \ . \tag{14.2}$$

Then subtract Eq. (14.2) from Eq. (14.1) using:

$$\begin{aligned}
\mathbf{E}\cdot(\nabla\times\mathbf{B}) - \mathbf{B}\cdot(\nabla\times\mathbf{E}) &= \varepsilon_{ijk}E_i\partial_j B_k - \varepsilon_{kji}B_k\partial_j E_i \quad \text{(choice of indices NB)} \\
&= \varepsilon_{ijk}\left[E_i\partial_j B_k + B_k\partial_j E_i\right] \quad (\text{as } \varepsilon_{kji} = -\varepsilon_{ijk}) \\
&= \varepsilon_{ijk}\partial_j[E_i B_k] = -\varepsilon_{jik}\partial_j E_i B_k \\
&= -\nabla\cdot[\mathbf{E}\times\mathbf{B}]
\end{aligned}$$

And:

$$\mathbf{E}\cdot\frac{1}{\mathrm{c}}\frac{\partial\mathbf{E}}{\partial t} = \frac{1}{2}\frac{1}{\mathrm{c}}\frac{\partial}{\partial t}E^2$$

and similarly for $\mathbf{B}$. Then:

$$\begin{aligned}
-\nabla\cdot[\mathbf{E}\times\mathbf{B}] &= \frac{1}{2\mathrm{c}}\frac{\partial}{\partial t}(E^2+B^2) + \frac{4\pi}{\mathrm{c}}\mathbf{E}\cdot\mathbf{j} \\
\Rightarrow \quad \frac{\partial}{\partial t}\left(\frac{E^2+B^2}{8\pi}\right) + \nabla\cdot\mathbf{S} &= -\mathbf{E}\cdot\mathbf{j} \ ,
\end{aligned}$$

where:

$$\mathbf{S} = \frac{\mathrm{c}}{4\pi}\mathbf{E}\times\mathbf{B} \quad \text{(the Poynting vector)} \tag{14.3}$$

14.2 Interpretation via Integration

Thus returning to the integrated form over a (time-independent) volume V:

$$\int \mathrm{d}V \frac{\partial}{\partial t}\left(\frac{E^2+B^2}{8\pi}\right) + \int \mathrm{d}V \nabla\cdot\mathbf{S} + \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}V \mathcal{E}_{\mathrm{ch}} = 0$$

$$\Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\left\{\int \mathrm{d}V\left(\frac{E^2+B^2}{8\pi}\right) + \mathcal{E}^{\mathrm{ch}}\right\}}_{\text{energy of field + charges (inside } V)} + \underbrace{\int \mathrm{d}\mathbf{A}\cdot\mathbf{S}}_{\text{flux of energy through boundary}} = 0 \ .$$

The interpretation is the rate of change of the field and particle energy inside V is equal to the flux of energy through the boundary.

14.3 Covariant Form: Toward $T^{\mu\nu}$

Now let us cast this equation (**not** integrated over volume) in covariant form:

$$\frac{\partial}{\partial t}\left(\frac{E^2+B^2}{8\pi}\right) + \nabla\cdot\mathbf{S} = -\mathbf{E}\cdot\mathbf{j} \ . \tag{14.4}$$

Now:

$$-\mathbf{E}\cdot\mathbf{j} = -F^{0k}j_k = -F^{0\nu}j_\nu \quad (\text{as } F^{00}=0) \tag{14.5}$$

Now the LHS. of Eq. (14.4) looks like "$\partial_\mu t^\mu$", with the proposed four-vector

$$t^\mu = \left(\frac{E^2+B^2}{8\pi}, \mathbf{S}\right) \ .$$

But, noting the 0 index in Eq. (14.5), we expect correct generalisation is the 0-component of the four-divergence of a tensor,

$$\partial_\mu T^{\mu 0} \propto -F^{0\nu}j_\nu \ ,$$

implying the general result

$$\partial_\mu T^{\mu\alpha} = -\frac{1}{\mathrm{c}}F^{\alpha\mu}j_\mu \tag{14.6}$$

where we chose the coefficient of proportionality to be $1/\mathrm{c}$ to obtain correct dimensional content.

We need an identity to establish the form of $T^{\mu\nu}$:

$$\partial_\mu {}^*F^{\nu\mu} = 0 \quad \Leftrightarrow \quad \partial_\mu F_{\nu\kappa} + \partial_\nu F_{\kappa\mu} + \partial_\kappa F_{\mu\nu} = 0 \ , \tag{14.7}$$

see problem (4.1).

Turning to the derivation of $T^{\mu\nu}$, consider the RHS. of Eq. (14.6) above and noting order of indices in F:

$$\partial_\nu F^{\nu\mu} = -\frac{4\pi}{\mathrm{c}}j^\mu \Rightarrow \quad \frac{1}{4\pi}\partial^\nu F_{\nu\mu} = -\frac{1}{\mathrm{c}}j_\mu \ . \tag{14.8}$$

implying, using Eq. (14.6) and moving the $1/4\pi$ to the LHS, and obtaining same index on ∂^μ on both sides:

$$\begin{aligned} 4\pi\partial_\mu T^{\mu\alpha} &= F^{\alpha\mu}\partial^\nu F_{\nu\mu} \\ \text{relabel } \mu\leftrightarrow\nu \quad &= F^{\alpha\nu}\partial^\mu F_{\mu\nu} \\ &= \partial^\mu(F^{\alpha\nu}F_{\mu\nu}) - (\partial^\mu F^{\alpha\nu})F_{\mu\nu} \ . \end{aligned} \tag{14.9}$$

Let us focus on the second term in Eq. (14.9)

$$
\begin{aligned}
-F_{\mu\nu}\partial^\mu F^{\alpha\nu} &= -\frac{1}{2}\left(F_{\mu\nu}\partial^\mu F^{\alpha\nu} - F_{\mu\nu}\partial^\mu F^{\nu\alpha}\right) \\
\text{(Relabel interchange } \mu \text{ and } \nu.) \quad &= -\frac{1}{2}\left(F_{\mu\nu}\partial^\mu F^{\alpha\nu} - F_{\nu\mu}\partial^\nu F^{\mu\alpha}\right) \\
\text{(Use antisymmetry } F) \quad &= -\frac{1}{2}\left(F_{\mu\nu}\partial^\mu F^{\alpha\nu} + F_{\mu\nu}\partial^\nu F^{\mu\alpha}\right) \\
\text{(Add/subtract extra term)} \quad &= -\frac{1}{2}F_{\mu\nu}\underbrace{\left(\partial^\mu F^{\alpha\nu} + \partial^\nu F^{\mu\alpha} + \partial^\alpha F^{\nu\mu}\right)}_{=0 \text{ from problem (4.1)}} + \frac{1}{2}F_{\mu\nu}\partial^\alpha F^{\nu\mu} \\
&= \frac{1}{2}F_{\mu\nu}\partial^\alpha F^{\nu\mu} \\
\text{(Duplicate term)} \quad &= \frac{1}{4}\left(F_{\mu\nu}\partial^\alpha F^{\nu\mu} + F_{\mu\nu}\partial^\alpha F^{\nu\mu}\right) \\
\text{(Flip-flop RH term.)} \quad &= \frac{1}{4}\left(F_{\mu\nu}\partial^\alpha F^{\nu\mu} + F^{\mu\nu}\partial^\alpha F_{\nu\mu}\right) \\
\text{(Use antisymmetry)} \quad &= -\frac{1}{4}\left(F_{\mu\nu}\partial^\alpha F^{\mu\nu} + F^{\mu\nu}\partial^\alpha F_{\mu\nu}\right) \\
&= -\frac{1}{4}\partial^\alpha\left[F_{\mu\nu}F^{\mu\nu}\right] \ . \qquad (14.10)
\end{aligned}
$$

Returning to Eq. (14.9), we see that Eq. (14.10) imples

$$
\begin{aligned}
4\pi\partial_\mu T^{\mu\alpha} &= \partial^\mu\left(F^{\alpha\nu}F_{\mu\nu}\right) - \frac{1}{4}\partial^\alpha\left(F_{\mu\nu}F^{\mu\nu}\right) \\
&= \partial^\mu\left(F^{\alpha\nu}F_{\mu\nu} - \frac{1}{4}\delta^\alpha_\mu F_{\lambda\nu}F^{\lambda\nu}\right) \\
\Rightarrow \quad &= \eta^{\mu\beta}\partial_\beta\left(F^{\alpha\nu}F_{\mu\nu} + \frac{1}{4}\delta^\alpha_\mu F_{\mu\nu}F^{\mu\nu}\right) \\
\text{(Relabel LHS } \mu \to \beta.) \quad \Rightarrow \quad T^{\beta\alpha} &= -\frac{1}{4\pi}\left(\eta^{\beta\mu}F^{\nu\alpha}F_{\mu\nu} + \frac{1}{4}\eta^{\alpha\beta}F_{\lambda\nu}F^{\lambda\nu}\right) \ . \qquad (14.11)
\end{aligned}
$$

Notes:

- The choice is not unique, as there is a "constant of integration" from integrating $\partial_\mu T^{\mu\nu}$, so we may choose:

$$T'^{\alpha\beta} = T^{\alpha\beta} + \Sigma^{\alpha\beta} \qquad (14.12)$$

  where $\partial_\mu \Sigma^{\alpha\mu} = 0$, and, when integrated, provides the constant of integration.

- Find T is symmetric. Note that the second term in the expression Eq. (14.11) is immediately symmetric as it is a scalar multiplying the metric tensor. So focus on the first term in Eq. (14.11), we must show it is symmetric in α and β:

$$
\begin{aligned}
\eta^{\beta\mu}F^{\nu\alpha}F_{\mu\nu} &= F^{\nu\alpha}F^\beta{}_\nu \\
\text{(Flip-flop on } \nu.) \quad &= F_\nu{}^\alpha F^{\beta\nu} \\
\text{(Use } F^{\beta\nu} = -F^{\nu\beta}) \quad &= -F_\nu{}^\alpha F^{\nu\beta} \\
&= -\eta^{\alpha\mu}F_{\nu\mu}F^{\nu\beta} \\
&= +\eta^{\alpha\mu}F_{\mu\nu}F^{\nu\beta} \ ,
\end{aligned}
$$

  establishing the symmetric nature of the first term (under $\alpha \leftrightarrow \beta$) in Eq. (14.11), for our choice of $\Sigma^{\beta\alpha} = 0$.

  Hence $T^{\alpha\beta}$ is symmetric, using our choice of the "constant of integration".

14.4 Components of $T^{\alpha\beta}$ in 3-Form

We will now see that the components of $T^{\mu\nu}$ have a clear physical meaning. We will "partition" the tensor in a similar manner to that used in considering $F^{\mu\nu}$. Since $T^{\mu\nu}$ is *symmetric*, we have *three* categories, T^{00}, T^{0i} and T^{ij}. We examine these in turn and interpret them.

T^{00} component:

$$\begin{aligned} -4\pi T^{00} &= \left(\eta^{0\mu}F^{\nu 0}F_{\mu\nu} + \frac{1}{4}\eta^{00}F_{\lambda\nu}F^{\lambda\nu}\right) \\ \text{(Use } \eta^{0\mu} = -\delta^{\mu}_{0}.) \quad &= \left(-F^{i0}F_{0i} - \frac{1}{4}F_{\lambda\nu}F^{\lambda\nu}\right) \\ &= -(-E^i)(-E^i) - \frac{1}{4}\cdot 2(B^2 - E^2) \\ &= -\frac{1}{2}(E^2 + B^2)\ . \end{aligned} \tag{14.13}$$

i.e. T^{00} is proportional to the **energy density**, w:

$$T^{00} = \frac{E^2 + B^2}{8\pi} = w \tag{14.14}$$

T^{0i} component:

$$\begin{aligned} -4\pi T^{0i} &= \eta^{0\mu}F^{\nu i}F_{\mu\nu} + \frac{1}{4}\underbrace{\eta^{0i}}_{=0}F^{\mu\nu}F_{\mu\nu} \\ &= -F^{\nu i}F_{0\nu} + 0 \\ &= -F^{ji}F_{0j} \\ &= -\varepsilon^{jik}B_k(-E_j) \\ &= -\varepsilon^{ijk}B_kE_j = -(\mathbf{E}\times\mathbf{B})_i \end{aligned} \tag{14.15}$$

So we find the Poynting vector, $\mathbf{S}$:

$$T^{0i} = \frac{1}{4\pi}(\mathbf{E}\times\mathbf{B})^i = \frac{1}{\mathrm{c}}S^i \tag{14.16}$$

So $\mathbf{S}$ is proportional to both the flux of energy and the momentum density.

T^{ij} component: For brevity we denote $F^{\mu\nu}F_{\mu\nu} \to FF$:

$$\begin{aligned} -4\pi T^{ij} &= \eta^{i\mu}F^{\nu j}F_{\mu\nu} + \frac{1}{4}\eta^{ij}FF \\ &= F^{\nu j}F_{i\nu} + \frac{1}{4}\delta^{ij}FF \\ &= F^{0j}F_{i0} + F^{kj}F_{ik} + \frac{1}{4}\delta^{ij}FF \\ &= E^iE^j + \varepsilon^{kj\ell}B_\ell\varepsilon^{ikm}B_m + \frac{1}{4}\delta^{ij}FF \\ &= E^iE^j + (\delta_{jm}\delta_{\ell i} - \delta_{ji}\delta_{\ell m})B_\ell B_m + \frac{1}{4}\delta^{ij}FF \\ &= E^iE^j + B^iB^j - \frac{1}{2}\delta^{ij}(B^2 + E^2) \end{aligned} \tag{14.17}$$

This is the (non-covariant) "**Maxwell stress tensor**". This represents the flux of momentum in the electromagnetic field – hence there are two indices to label surface normal and the component of momentum. This interpretation will become clearer below,

So in summary:

$$T^{00} = \frac{E^2 + B^2}{8\pi} = w$$

$$T^{0i} = \frac{1}{4\pi}(\mathbf{E} \times \mathbf{B})^i = \frac{1}{\mathrm{c}} S^i$$

$$T^{ij} = -\frac{1}{4\pi}\left\{E^i E^j + B^i B^j - \frac{\delta^{ij}}{2}(E^2 + B^2)\right\}$$

14.5 Connection with Momentum Conservation

To see the connection with momentum conservation of fields and particles, integrate:

$$\partial_\mu T^{\mu\beta} = -\frac{1}{\mathrm{c}} F^{\beta\nu} j_\nu \tag{14.18}$$

over volume V:

$$\int \mathrm{d}V \partial_\mu T^{\mu\beta} = -\frac{1}{\mathrm{c}} \int \mathrm{d}V F^{\beta\nu} j_\nu. \tag{14.19}$$

Consider $\beta = i$, the spatial part. ($\beta = 0 \Rightarrow$ energy conservation, as before.)

Since $j_\nu = \rho_0 u_\nu$, we see the RHS:

$$-\frac{1}{\mathrm{c}} \int \mathrm{d}V \rho_0 F^{i\nu} u_\nu \ . \tag{14.20}$$

But the integrand is the RHS of the Lorentz equation:

$$-\frac{1}{\mathrm{c}} \int \mathrm{d}V \rho_0 F^{i\nu} u_\nu = -\frac{\mathrm{d}p^i}{\mathrm{d}t}^{\text{part}} \ . \tag{14.21}$$

Now consider the LHS:

$$\int \mathrm{d}V \{\partial_0 T^{0i} + \partial_j T^{ji}\} \ . \tag{14.22}$$

Assuming the volume of integration is **not** time-dependent,

$$\frac{1}{\mathrm{c}} \int \frac{\partial T^{0i}}{\partial t} \mathrm{d}V + \frac{\mathrm{d}p_i^{\text{part}}}{\mathrm{d}t} = -\int \mathrm{d}V \partial_j T^{ji}$$

$$\Rightarrow \quad \frac{1}{\mathrm{c}} \frac{\mathrm{d}}{\mathrm{d}t} \int T^{0i} \mathrm{d}V + \frac{\mathrm{d}p_i^{\text{part}}}{\mathrm{d}t} = -\int \mathrm{d}V \partial_j T^{ji}$$

(Use divergence theorem.) $$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\frac{1}{\mathrm{c}} \int T^{0i} \mathrm{d}V + p_i^{\text{part}}\right\} = -\int \mathrm{d}S_j T^{ji} \ .$$

Thus if $T^{ji} \to 0$ fast enough on a large surface, so that no momentum is lost to radiation from the volume, it is clear that:

$$\frac{1}{\mathrm{c}} \int \mathrm{d}V T^{0i} = p_i^{\text{field}}$$

and the expression of overall momentum conservation for field and particles:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(p_i^{\text{field}} + p_i^{\text{part}}\right) = 0 \ .$$

The significance of the surface integral which we discarded is thus clear:

$$-\int \mathrm{d}S_j \, T^{ji} \tag{14.23}$$

is the **flux of momentum** out of the volume. Thus the equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(p_i^{\text{field}} + p_i^{\text{part}}\right) = -\int \mathrm{d}S_j \, T^{ji} \tag{14.24}$$

expresses overall momentum conservation.

Eq. (14.24) is the counterpart to conservation equations for mass or charge from the ordinary continuity equation. Here Σ is the boundary of the volume V:

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= -\nabla \cdot \mathbf{j} \\ \Rightarrow \quad \int_V \mathrm{d}V \, \frac{\partial \rho}{\partial t} &= -\int_V \mathrm{d}V \, \nabla \cdot \mathbf{j} \\ \Rightarrow \quad \frac{\mathrm{d}Q}{\mathrm{d}t} &= -\int_\Sigma \mathrm{d}\mathbf{S} \cdot \mathbf{j} \ , \end{aligned}$$

leading to the rate of change of the total charge, Q, in the (time-independent) volume V being equal to the flux through the surface Σ. Q is replaced by the total momentum of the particles and field and $\mathbf{j}$ is replaced by the energy-momentum tensor, T^{ij}, reflecting the transport of momentum across the boundary, Σ.